Ex 1:

1) We show that:

$\zeta + \frac{1}{\alpha}\mathbb{E}[(-\Delta V(x) - \zeta)^+]$ is convex

Proof:

$$lCVaR(\alpha, \lambda x + (1-\lambda)x_2, \mu\zeta_1 + (1-\mu)\zeta_2) \leq \lambda lCVaR(\alpha, x_1, \zeta_1) + (1-\lambda) lCVaR(\alpha, x_2, \zeta_2)$$

$$= \lambda\zeta_1 + (1-\lambda)\zeta_2 + \frac{1}{\alpha}\mathbb{E}[(-V(x + \lambda x_1^T(R - r\mathbb{1}) + (1-\lambda)x_2^T(R - r\mathbb{1})) - \lambda\zeta_1 - (1-\lambda)\zeta_2]$$

$$\leq \lambda\zeta_1 + (1-\lambda)\zeta_2 + \frac{1}{\alpha}\mathbb{E}[(-V(1 + \lambda x_1^T(R - r\mathbb{1})) - V(1 + (1-\lambda)x_2^T(R - r\mathbb{1})) - \lambda\zeta_1 - (1-\lambda)\zeta_2]$$

$$\leq \lambda\zeta_1 + \frac{1}{\alpha}\mathbb{E}[\lambda(-V(1 + x_1^T(R - r\mathbb{1})) - \zeta_1)^+ + (1-\lambda)(-V(1 + x_2^T(R - r\mathbb{1})) - (1-\lambda)\zeta_2)^+]$$

$$\leq \lambda\zeta_1 + \frac{\lambda}{\alpha}\mathbb{E}[(-V(1 + x_1^T(R - r\mathbb{1})) - \zeta_1)^+] + (1-\lambda)\zeta_2 + \frac{1-\lambda}{\alpha}\mathbb{E}[(-V(1 + x_2^T(R - r\mathbb{1})) - \zeta_2)^+]$$

$$\leq \lambda\, lCVaR(\alpha, x_1, \zeta_1) + (1-\lambda)\, lCVaR(\alpha, x_2, \zeta_2)$$

Convex.

b) We showed it's convex wrt $(x, \gamma) \in \mathbb{R}^2$ $\Rightarrow$ its convex for $(\gamma) \in \mathbb{R}$.

- Continuously differentiable?

$$\frac{\partial}{\partial \gamma} \ell CVaR = 1 + \frac{1}{\alpha} \mathbb{E}\left[\frac{\partial}{\partial \gamma} \max(-\Delta V - \gamma, 0)\right]$$

$$\ell CVaR = \begin{cases} \cdot\ \gamma + \frac{1}{\alpha} \mathbb{E}[-\Delta V - \gamma] & \text{if } -\Delta V - \gamma \geq 0 \\ & \Rightarrow -\Delta V \geq \gamma. \\ \cdot\ \gamma + \frac{1}{\alpha} \mathbb{E}[0] = \gamma \end{cases}$$

$$\frac{\partial}{\partial \gamma} \ell CVaR = \begin{cases} 1 + \frac{1}{\alpha} \mathbb{E}[-1] = \left(1 - \frac{1}{\alpha}\right), \text{ if } -\Delta V \geq \gamma. \\ 1 \qquad \text{else} \end{cases}$$

$$= 1 - \frac{1}{\alpha} \mathbb{1}_{\{-\Delta V \geq \gamma\}}$$

Or else:

$$\frac{\partial}{\partial \varphi} \ell CVaR = 1 + \frac{1}{\alpha} \mathbb{E}\left[-1 \, \mathbb{1}_{\{-\Delta V \geq \varphi\}} \right]$$

$$= 1 - \frac{1}{\alpha} \mathbb{Q}(-\Delta V > \varphi)$$

f is convex and our objective function is continuously differentiable
a solution φ^*: can be found solving $\frac{\partial}{\partial \varphi} \ell(VaR(\alpha, x, \varphi^*) = 0$

$$1 - \frac{1}{\alpha} \mathbb{Q}(-\Delta V > \varphi^*) = 0$$

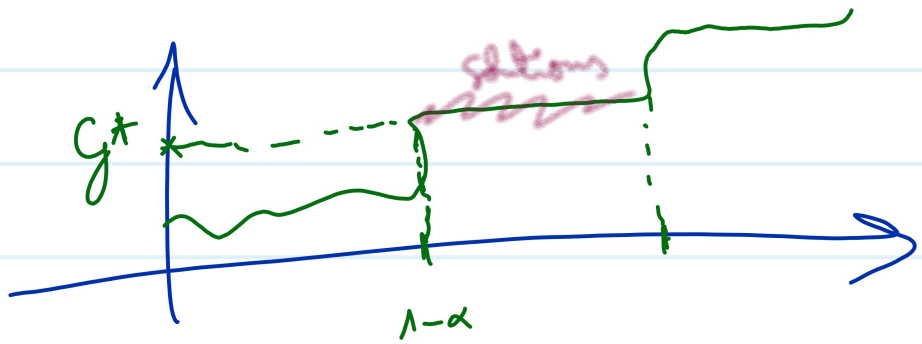
$$\mathbb{Q}(-\Delta V > \varphi^*) = \alpha$$

$$1 - F_{-\Delta V}(\varphi^*) = \alpha$$

$$\mathbb{Q}(-\Delta V \leq \varphi^*) = F_{-\Delta V}(\varphi^*) = 1 - \alpha$$

Since R is a continuous distribution $\Rightarrow -\Delta V(x)$ has a continuous distribution
there exists at least φ solution to the previous equation:

$$\Rightarrow \boxed{\varphi^* = F^{-1}_{-\Delta V}(1-\alpha)}$$



Left bound : $VaR_0(\alpha, x) = \inf\{G \in \mathbb{R} : \mathbb{Q}(-\Delta V(x) \leq G) \geq 1-\alpha\}$

$= \min\{G \in \mathbb{R} : \mathbb{Q}(-\Delta V(x) \leq G) = 1-\alpha\}$

$= \min_{G \in A} G^*$.

with A bounded set from above and below

Cuz this is a cdf.

c) $\ell CVaR(\alpha, x, VaR_0)$

$= G + \frac{1}{\alpha} \mathbb{E}[(-\Delta V(x) - VaR_0)^+] =$

$= G + \frac{1}{\alpha} \mathbb{E}[(-\Delta V(x) - VaR_0) \mathbb{1}_{\{-\Delta V(x) \geq VaR_0\}}]$

$= G + \frac{1}{\alpha}\left[\mathbb{E}[-\Delta V(x) \mathbb{1}_{\{-\Delta V \geq VaR_0\}}] - VaR_0 \mathbb{E}[\mathbb{1}_{\{-\Delta V \geq VaR_0\}}]\right]$

$$= \gamma + \frac{1}{\alpha}\left[- \text{VaR}_\alpha \, \mathbb{Q}(-\Delta V \geq \text{VaR}_\alpha) + \mathbb{E}[-\Delta V(x) \mid -\Delta V \geq \text{VaR}] \right]$$

$\mathbb{Q}(-\Delta V \leq \gamma^*) = 1 - \alpha$.

$\Rightarrow \mathbb{Q}(-\Delta V \geq \gamma^*) = \alpha$

$$= \gamma + \frac{1}{\alpha} \mathbb{E}[-\Delta V - \text{VaR}_\alpha \mid -\Delta V \geq \text{VaR}_\alpha] \, \mathbb{Q}(-\Delta V \geq \text{VaR}_\alpha)$$

$$= \text{VaR}_\alpha - \text{VaR}_\alpha + \mathbb{E}[-\Delta V \mid -\Delta V \geq \text{VaR}_\alpha].$$

c) Let's show solutions are equivalent

We add $\min_{x \in \mathbb{R}}$ on both sides.

- Let's assume: x^* solution of P_{CVaR}

we know $\gamma^* = \text{VaR}_\alpha(\alpha, x)$ is solution to P_{CVaR}.
$P_{CVaR}(x^*) = P_{CVaR}(\text{VaR}_\alpha(\alpha, x^*))$

$\Rightarrow \text{VaR}_\alpha(\alpha, x^*)$ is solution to P_{CVaR}.

$\Rightarrow \left(x^*, \text{VaR}_\alpha(\alpha, x^*)\right)$ is // P_{CVaR}.

- Let's assume $\left(x^*, \text{VaR}_\alpha(\alpha, x^*)\right)$ solution to P_{CVaR}

then $\text{VaR}_\alpha(\alpha, x^*)$ solution to P_{CVaR}.

but $P_{CVaR}(\text{VaR}_\alpha(\alpha, x^*)) = P_{CVaR}(x^*)$

$\Rightarrow x^*$ is solution to P_{CVaR}.

$$\min_{x \in X} CVaR(\alpha, x) = \min_{\zeta \in \mathbb{R}, x \in X} lCVaR(\alpha, x, \zeta)$$

$\zeta = VaR_\alpha$

x^* sol to CVaR:

$$\min_{x \in X} CVaR(\alpha, x^*) = CVaR(\alpha, x^*) = \min lCVaR(\alpha, x^*, VaR_\alpha(\alpha, x^*))$$

thus $(x^*, VaR_\alpha(\alpha, x^*))$ solution to PlCVaR.

- $(x^*, VaR_\alpha(\alpha, x^*))$ sol to PlCVaR:

$$\min lCVaR(\alpha, x^*, VaR_\alpha(\alpha, x^*)) = CVaR(\alpha, x^*) = \min_{x \in X} CVaR(\alpha, x^*)$$